

Mathematics II (BSM 121)

Manju Subedi

Gandaki University
Bachelor in Information Technology (BIT)
BSM 121

manjusubedi.061@gmail.com

June 8, 2026

Outlines

- Modulus of Complex Numbers and its Properties
- Integer Powers of Z . De Moivre's Formula
- Square Roots of Complex Numbers
- n^{th} Roots of Complex Numbers

Modulus of Complex Numbers and its Properties

Modulus of complex number $z = x + iy$ is defined as

$$|z| = \sqrt{x^2 + y^2} = \sqrt{z\bar{z}}$$
$$\implies |z|^2 = z\bar{z}$$

Properties

① Triangle Inequality:

$$|z_1 + z_2| \leq |z_1| + |z_2|$$

$$|z_1 - z_2| \geq ||z_1| - |z_2||$$

$$|z_1 + z_2| \geq ||z_1| - |z_2||$$

② Generalized triangle inequality:

$$|z_1 + z_2 + \cdots + z_n| \leq |z_1| + |z_2| + \cdots + |z_n|$$

③ Product: $|z_1 z_2| = |z_1| |z_2|$

④ Division: $\left| \frac{z_1}{z_2} \right| = \frac{|z_1|}{|z_2|}$

Example: If $z_1 = 3 + 4i$ and $z_2 = 2 + i$ then verify triangle inequality properties, product property and division property of modulus of complex numbers.

Integer Powers of z . De Moivre's Formula

Euler's Formula: $e^{i\theta} = \cos \theta + i \sin \theta$

$z^n = [re^{i\theta}]^n = r^n e^{in\theta} = r^n (\cos n\theta + i \sin n\theta)$, Where $n = 0, 1, 2, \dots$

De Moivre's formula: For any positive integer n ,

$$[r(\cos \theta + i \sin \theta)]^n = r^n (\cos n\theta + i \sin n\theta)$$

n^{th} Roots of Complex Numbers: If w be any non-zero complex number and n is positive integer such that $z^n = w$, then z is called n^{th} root of complex number w . It is denoted by

$$z = w^{1/n} \text{ or } \sqrt[n]{w}.$$

Let, $w = r(\cos \theta + i \sin \theta)$, then we may write
 $w = r[\cos(\theta + 2\pi k) + i \sin(\theta + 2\pi k)]$, where k is a whole number.

Continue...

$$\begin{aligned}\text{Now, } w^{1/n} = z_k &= [r(\cos(\theta + 2\pi k) + i \sin(\theta + 2\pi k))]^{\frac{1}{n}} \\ &= r^{1/n} \left[\cos \frac{(\theta + 2\pi k)}{n} + i \sin \left(\frac{\theta + 2\pi k}{n} \right) \right]\end{aligned}$$

Using De-moivre's theorem, where, $k = 0, 1, 2, \dots, n-1$ Then z_k is n^{th} root of ω for $k = 0, 1, 2, \dots, n-1$.

Example

Use De-Moivre's theorem to evaluate cube roots of unity.

Solution:

Here, $z = 1 = x + 0i \implies \theta = 0^\circ$ and, $r = 1$

then, $z = \cos 0^\circ + i \sin 0^\circ = \cos(k \cdot 360^\circ + 0) + i \sin(k \cdot 360^\circ + 0)$
 $= \cos k \cdot 360^\circ + i \sin k \cdot 360^\circ.$

Then, $z^{1/3} = z_k = [\cos k \cdot 360^\circ + i \sin k \cdot 360^\circ]^{1/3}$

So, when

$$k = 0, \quad z_0 = \cos 0 + i \sin 0 = 1 + i \cdot 0 = 1$$

$$k = 1, \quad z_1 = \cos \frac{360^\circ}{3} + i \sin \frac{360^\circ}{3} = \cos 120^\circ + i \sin 120^\circ = -\frac{1}{2} + i \frac{\sqrt{3}}{2}$$

$$k = 2, \quad z_2 = \cos \frac{720^\circ}{3} + i \sin \frac{720^\circ}{3} = \cos 240^\circ + i \sin 240^\circ = -\frac{1}{2} - i \frac{\sqrt{3}}{2}$$

$$\therefore \sqrt[3]{1} = 1, \frac{-1+i\sqrt{3}}{2}, \frac{-1-i\sqrt{3}}{2}$$

Exercise: Use De-Moivre's theorem to find the 5^{th} and 8^{th} roots of unity

Properties of Cube roots of Unity

Let ω and ω^2 be imaginary roots of unity, then we have

- Each imaginary cube root of unity is square of the other.
i.e. $\omega^2 = \omega$
- The product of two imaginary cube roots of unity is 1.
i.e. $\omega \cdot \omega^2 = \omega^3 = 1$
- Sum of all the cube roots of unity is zero.
i.e. $1 + \omega + \omega^2 = 0$

If ω and ω^2 are imaginary roots of unity, then

- (i) Verify all the properties of cube roots of unity.
- (ii) Use above properties to show $(1 - \omega + \omega^2)(1 + \omega - \omega^2) = 4$

Square Roots of Complex Numbers

Consider a complex number $z = a + ib$

Let, $x + iy$ be the square root of $a + ib$

i.e. $x + iy = \sqrt{a + ib}$

Then, $(x + iy)^2 = a + ib$

or, $x^2 + 2ixy + i^2y^2 = a + ib$

or, $(x^2 - y^2) + i.2xy = a + ib$

Equating real and imaginary parts we get, $x^2 - y^2 = a \dots\dots (i)$ and

$2xy = b \dots\dots (ii)$

Now, $(x^2 + y^2)^2 = (x^2 - y^2)^2 + 4x^2y^2 = a^2 + b^2$

$\therefore x^2 + y^2 = \sqrt{a^2 + b^2} \dots\dots (iii)$

Adding (i) and (iii) we get,

$$2x^2 = a + \sqrt{a^2 + b^2} \implies x^2 = \frac{a + \sqrt{a^2 + b^2}}{2}$$

$$\therefore x = \pm \left(\frac{a + \sqrt{a^2 + b^2}}{2} \right)^{\frac{1}{2}}$$

Square Roots of Complex Numbers Continue

Again, subtracting (i) from (iii) we get,

$$\begin{aligned}2y^2 &= \sqrt{a^2 + b^2} - a \\ \text{or, } y^2 &= \frac{\sqrt{a^2 + b^2} - a}{2} \\ \text{or, } y &= \pm \left(\frac{\sqrt{a^2 + b^2} - a}{2} \right)^{\frac{1}{2}}\end{aligned}$$

$\therefore$ The required square roots of $a + ib$ are $x + iy$, where

$$x = \pm \left(\frac{a + \sqrt{a^2 + b^2}}{2} \right)^{\frac{1}{2}} \text{ and } y = \pm \left(\frac{\sqrt{a^2 + b^2} - a}{2} \right)^{\frac{1}{2}}$$

Example

Example: Find the square roots of $z = -8 + 6i$

Solution: let $a + ib$ be the square roots of $z = -8 + 6i = x + iy$ then,

$$\begin{aligned}a &= \pm \left(\frac{x + \sqrt{x^2 + y^2}}{2} \right)^{1/2} = \pm \left(\frac{-8 + \sqrt{64 + 36}}{2} \right)^{1/2} \\&= \pm \left(\frac{-8 + 10}{2} \right)^{1/2} = \pm 1^{1/2} = \pm 1\end{aligned}$$

$$\begin{aligned}b &= \pm \left(\frac{\sqrt{x^2 + y^2} - x}{2} \right)^{1/2} = \pm \frac{\sqrt{64 + 36} + 8}{2}^{1/2} \\&= \pm \left(\frac{10 + 8}{2} \right)^{1/2} = \pm 3\end{aligned}$$

$\therefore \pm(1 + 3i)$ is the required square roots of given complex number

Exercise 1. Find the square roots of (i) $5 - 12i$ (ii) $7 - 24i$

Thank You